

PROJECT REVIEW – 2

Explainable Vision-Language Diagnostic Assistant

for Chest Radiographs with Offline Rural Deployment

NLP Specialization Capstone Project

P. Kavya Sai – 2420030322

D. Poojitha – 2420030697

P. Meghana – 2420030401

S. Anajana – 2420030499

Cluster – 3

Section-7

Faculty: Dr. K Swanthana

Problem Statement

BACKGROUND & CHALLENGE

- 👤 **Acute Radiologist Shortage:** Primary health centers in rural regions lack specialist radiologists to read X-rays and detect conditions like pneumonia or tuberculosis.
- 🕒 **Diagnostic Delays & Travel Burden:** Patients travel vast distances to urban hospitals, delaying time-critical care and incurring heavy financial expense.
- 👁️ **Black-Box AI Mistrust:** Traditional AI systems provide class outputs without explanations, making local nurses reluctant to rely on machine predictions.





1. Multimodal Diagnosis

Develop a vision-language assistant to generate plain-language diagnostic descriptions (e.g., "early signs of pneumonia") alongside class confidence scores.



2. Visual Explanation

Incorporate Grad-CAM heatmaps and attention overlays that circle the exact anatomical regions of concern, giving healthcare workers verifiable proof.



3. Automated Triage & Priority Guidance

Compute case urgency flags (e.g., "Please refer to specialist within 24 hours") so rural health workers can prioritize critical emergency cases immediately.

Literature Survey & Research Gap

Research Work / Model	Key Contribution	Limitation / Research Gap	Research Paper
ResNet & DenseNet	Strong CNN-based image classification and feature extraction.	Mainly focused on classification; limited report generation and multimodal reasoning.	ResNet Paper / DenseNet Paper
CheXNet	Detects pneumonia and other thoracic diseases from chest X-rays using deep CNNs.	Primarily classification-based and lacks comprehensive multimodal reasoning.	CheXNet Paper
Grad-CAM	Generates heatmaps showing image regions responsible for model predictions.	Visual explanation is not inherently connected to textual diagnosis or clinical reasoning.	Grad-CAM Paper
BLIP-2	Connects image encoders with large language models for vision-language understanding and generation.	Large language models can still require substantial computational resources.	BLIP-2 Paper
MedKLIP	Incorporates medical knowledge and aligns medical entities with visual image regions for X-ray diagnosis.	More complex architecture and computational requirements make lightweight deployment difficult.	MedKLIP Paper

Literature Survey & Research Gap

Research Work / Model	Key Contribution	Limitation / Research Gap	Research Paper
GLoRIA	Learns global and local image-text representations by associating image regions with words from radiology reports.	Requires paired image-report data and does not directly provide actionable triage recommendations.	GLoRIA Paper
ConVIRT	Uses paired medical images and text for contrastive learning and improves medical image representation with limited labels.	Focuses mainly on representation learning rather than complete diagnosis, explanation and triage.	ConVIRT Paper
MedCLIP	Uses medical knowledge-based contrastive learning with unpaired medical images and text.	Primarily supports representation learning and retrieval/classification rather than report generation and triage.	MedCLIP Paper
ChestX-ray8 / ChestX-ray14	Provides large-scale chest X-ray datasets for disease classification and weakly supervised localization.	Labels are largely derived from reports and the datasets alone do not provide multimodal clinical reasoning.	ChestX-ray8 Paper
Standalone Saliency Maps	Highlight important regions of medical images to support visual interpretation.	Usually lack textual diagnostic explanation, severity assessment and actionable triage guidance.	Grad-CAM Reference Paper

Innovation, Creativity & Novelty



Dual-Stream VLM Core

Combines Vision Transformers (ViT) for spatial feature extraction, CLIP for vision-text alignment, and BLIP-2 for diagnostic report synthesis into a unified framework.



Offline Edge Deployment

Pioneers 8-bit quantization on multimodal medical networks, enabling complete offline execution on mid-range Android mobile devices in zero-connectivity posts.



Transparent Reasoning

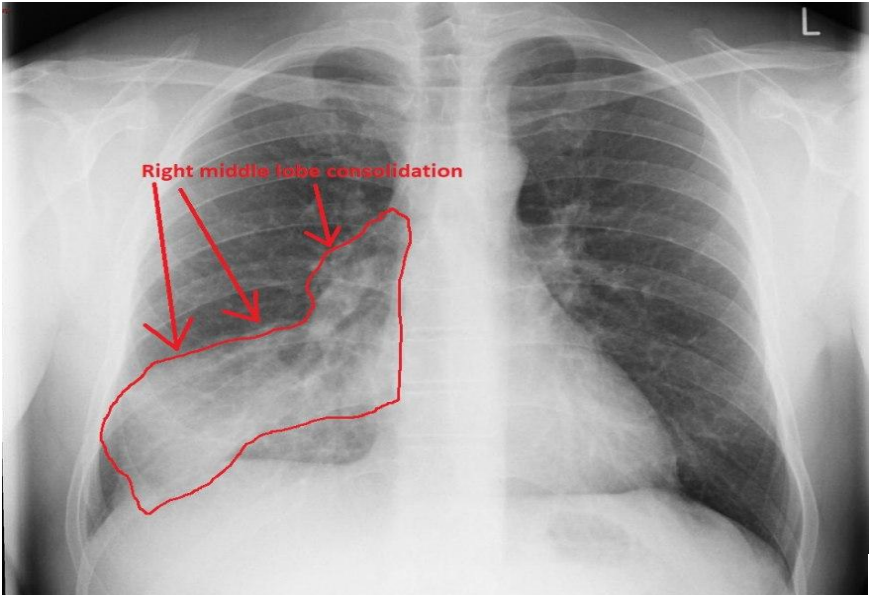
Generates dual Grad-CAM heatmaps and cross-attention overlays on the X-ray scan, bridging the trust gap between automated AI outputs and rural clinical workers.

Training Datasets

Dataset	Resource URL	Key Details
ChestX-ray14	NIH CC Box	112,120 chest X-rays with 14 disease labels.
OpenI (OpenAccess)	NLM OpenI	7,470 X-rays paired with clinical radiology reports.
ROCO Medical	GitHub / ROCO	Multimodal radiology dataset for text-image alignment.

Pipeline Flow

-  **Multi-Source Preprocessing:** Normalization & augmentations.
-  **ViT + CLIP Alignment:** Encodes spatial features into shared space.
-  **Dual Heads:** Disease Classifier + BLIP-2 Natural Text Generator.
-  **Explainability Layer:** Grad-CAM heatmaps & attention maps.
-  **Android Runtime:** Quantized model providing offline triage.



Methodology

DS

Data-Driven Class Selection

11 of 14 NIH conditions retained, chosen by measured per-class support

TL

Frozen-Backbone Transfer Learning

ImageNet-pretrained ViT-B/32 & DenseNet121; only final blocks fine-tuned

CB

Imbalance-Aware Training

Weighted loss + best-checkpoint-by-F1 to counter overfitting

TT

Post-Hoc Threshold Tuning

Per-class PR-curve optimization — no retraining required

RA

Retrieval-Augmented Reporting

TF-IDF grounding in real OpenI radiology report text

XC

Independent Cross-Checks

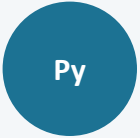
Zero-shot CLIP + BLIP v1 as secondary, separate signals

System Design



Two independent branches — vision explainability and text reporting — recombine at the presentation layer, alongside supplementary CLIP and BLIP v1 cross-checks.

Modern Tools and Technologies



PyTorch / torchvision

Model definition & training



**HuggingFace
Transformers**

CLIP & BLIP v1



scikit-learn

TF-IDF, PR-curves, metrics



OpenCV

Heatmap overlay & contours



pandas

Dataset & report handling



Streamlit

Interactive web app



fpdf2

Downloadable PDF reports



Kaggle / kagglehub

Dataset acquisition

Implementation

Pipeline Stages

Data → dataset.py, count_labels.py

Model → model.py, train.py

Explainability → gradcam.py

Evaluation → evaluate.py, tune_thresholds.py

Reporting → report_index.py, report_generator.py

Cross-checks → clip_verify.py, blip_caption.py

Decision → triage.py

Delivery → inference.py, report_pdf.py, app/app.py

Key Files & Purpose

model.py / train.py

ViT-B/32 transfer learning + imbalance-aware training

gradcam.py

Attention-based explainability, heatmap & contour marking

report_generator.py

Multi-condition, retrieval-grounded report text

triage.py

Deterministic urgency scoring

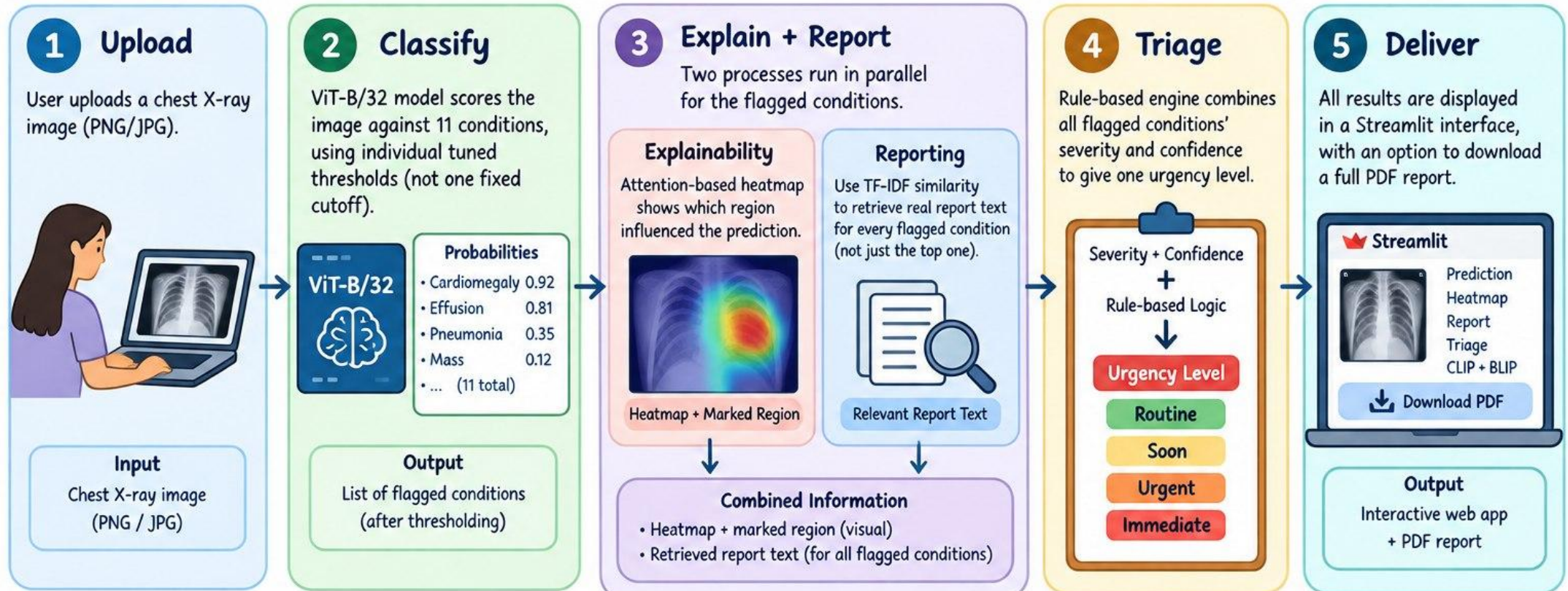
clip_verify.py / blip_caption.py

Independent CLIP & BLIP v1 cross-checks

app.py / report_pdf.py

Streamlit UI + downloadable PDF report

System Architecture



Key Technologies & Tools



PyTorch + torchvision
Model definition, training & fine-tuning



Hugging Face Transformers
CLIP (zero-shot) & BLIP-v1 (captioning)



scikit-learn
TF-IDF, cosine similarity, threshold tuning, evaluation metrics



OpenCV
Heatmap overlay, contour detection, region marking



pandas
Dataset & report handling



Streamlit
Web interface



fpdf2
PDF report generation



Kaggle + kagglehub
Dataset acquisition

Results

11

Conditions Classified

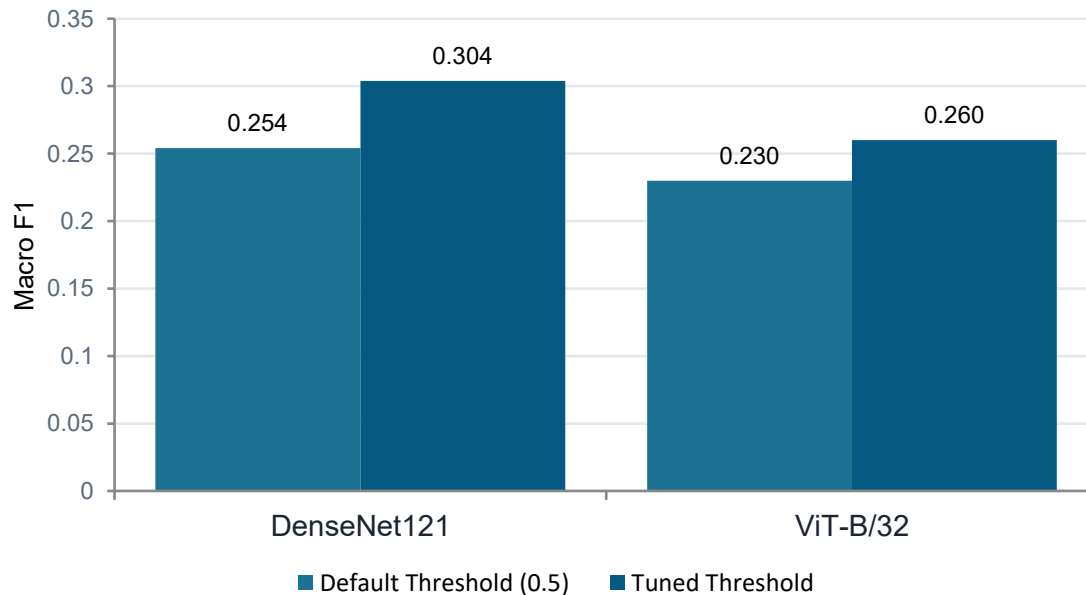
0.304

Best Macro F1 (tuned)

7,426

Real Reports Indexed

Backbone Comparison — Macro F1 Score

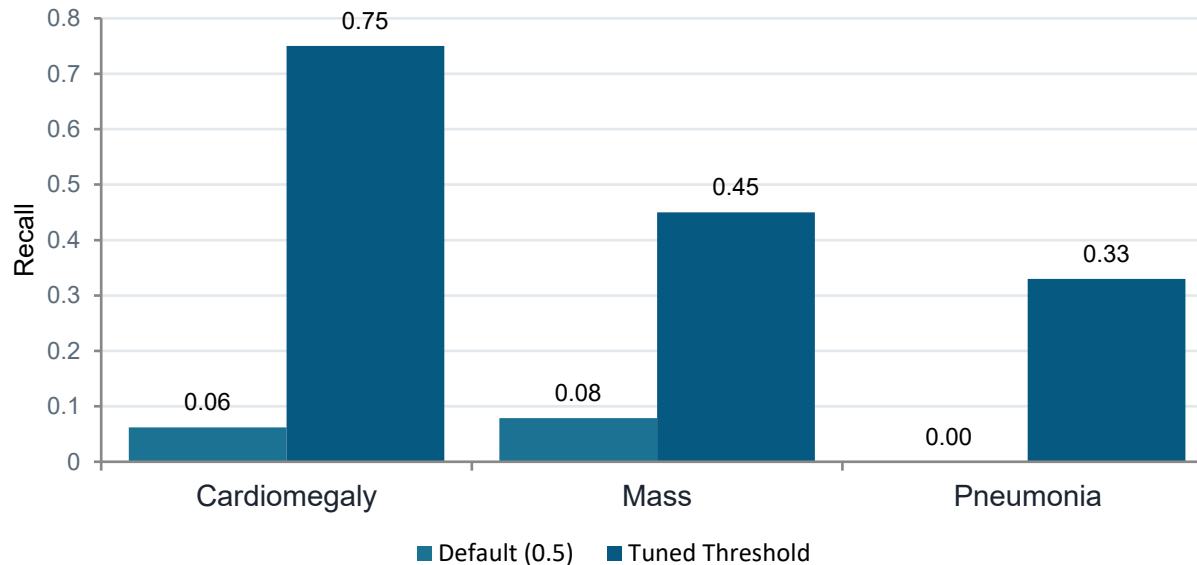


Why ViT scored slightly lower

Vision Transformers lack the built-in locality and translation-invariance biases CNNs have, so they typically need more training data to match CNN performance. With only ~4,800 training images, this gap is an expected, well-documented outcome — not an implementation defect.

Performance Evaluation

Recall Improvement from Per-Class Threshold Tuning



Trade-off accepted deliberately: for a triage tool, a missed finding is worse than a false alarm.

Qualitative validation: on a held-out case (ground truth: Cardiomegaly, Edema, Effusion), the system correctly flagged Cardiomegaly at 99.9% confidence and Effusion after the multi-condition reporting upgrade.

Explainability Validation

0.000 → 0.277

Heatmap std. deviation: gradient-based Grad-CAM (collapsed) vs. attention-based visualization (real signal)

Grad-CAM, adapted from CNNs, collapsed on this ViT due to token homogenization in the mostly-frozen backbone. Attention-based visualization — the standard ViT approach — restored genuine spatial signal.

Thank You

Questions & Discussion

Our project aims to bridge the rural diagnostic gap by delivering an honest, explainable, and fully offline AI assistant and safe care regardless of geographic limitations.